

ECONOMY — Agriculture and Allied Sector

Batch 8/10 · Lucent §4 (full) — Parmar has no agriculture chapter, Lucent-only + PYQ bucket "09 Agriculture and Food Management" (24 Qs) · ★=PYQ-tested · [B]=PYQ-book-only

1 Agriculture — Current Growth & Output

- Agri sector avg annual growth 4.6%(last 6yrs, as printed) → grew 3.0%(2021-22) vs 3.3%(2020-21)
- India rapidly emerging as net exporter of agri products → 2020-21 agri+allied exports +18% YoY; 2021-22 agri exports=all-time-high **US\$50.2bn**
- Agriculture&Allied = **18.4%** of India's GVA(current prices, 2022-23, BHARAT2025); ~1/7th of GDP; sustains ~2/3rd of population
- Real GVA grew 6.2%(H1 FY25); agri+services=key growth drivers; agri Q2 FY25=3.5%(best in 4 quarters) ← healthy Kharif production/above-normal monsoons/adequate reservoirs
- 1st advance estimates 2024-25: Kharif foodgrain=record **1647.05 LMT**(+5.7% vs2023-24, +8.2% vs5yr avg); rabi sowing(wheat+gram, as of10Jan2025)=+1.4%/+0.8% YoY
- Foodgrain production: 265.05mn tonnes(2013-14)→328.85mn tonnes(2023-24, record); horticulture=352.23mn tonnes(3rd advance estimate)

2 Millets, PMFBY & Commercial Crops

- International Year of Millets(IYM) → declared by UNGA(75th session, March2021) for 2023; Millets="Smart Food"(high nutrition, climate-resilient, aligned w/ UN SDGs)
- Commercial crops(def) → produced for trade not self-consumption: Oil-seeds, Sugar crops, Fibre crops, Narcotic crops, Beverage crops
- ACPC(Agriculture Cost&Price Commission) → decides MSP on govt's behalf
- MSP(def) → price at which govt is ready to purchase crop from farmers directly, if market price falls below MSP

HOOK PMFBY → launched **2016 Kharif season**; comprehensive crop-failure insurance for stabilising farmer income; world's **largest** crop-insurance scheme by farmer enrolments(~5.5cr applications/yr avg), **3rd-largest** by premium received; farmer pays only **1.5%(Rabi)/2%(Kharif)** of premium, Centre+State bear rest; 6yrs implementation: farmers paid ₹25,186cr premium, received claims ₹1.2 lakh cr(as of31Oct2022).

3 Sweet Revolution, Blue Economy 2.0 & Agri Boards

- Sweet Revolution → ₹500cr approved for National Beekeeping&Honey Mission(NBHM, 2020-21 to2022-23, part of ANB scheme); scientific beekeeping via National Bee Board(NBB)
- Blue Economy 2.0 → highlighted in Interim Budget 2024(1Feb2024); pillars=Mariculture/Maritime Transport/Renewable energy/Tourism; sustainable coastal-aquaculture/seas/coasts, climate-resilient; World Bank def="sustainable use of ocean resources for economic growth, improved livelihoods&jobs while preserving ocean-ecosystem health"

Board	HQ	Act
Coffee Board	Bengaluru(Karnataka)	Coffee Act,1942
Rubber Board	Kottayam(Kerala)	Rubber Act(Kerala),1947
Tea Board	Kolkata(WB)	Tea Act,1953
Tobacco Board	Guntur(AP)	Tobacco Act(AP),1975
Spices Board	Kochi(Kerala)	Spices Act,1986
National Meat&Poultry Processing Board	Delhi	26 Dec2008
Indian Grape Processing Board	Pune(Maharashtra)	2 Jan2009

4 Allied Sectors & NFSA

- Allied sectors(livestock/forestry&logging/fishing&aquaculture) → buoyant-growth sectors, potential for better farm incomes

- Livestock sector → CAGR **7.9%**(2014-15 to 2020-21, constant prices); contribution to agri-GVA ↑ 24.3% (2014-15)→30.1%(2020-21)
- Fisheries sector → annual avg growth ~7% since 2016-17; ~6.7% share of total agri GVA
- Dairy sector → most critical livestock component, employs >8cr farmers directly; India ranks **1st** in milk production, **3rd** in egg production, **8th** in meat production globally

HOOK NFSA 2013(notified 10.09.2013) → objective=food&nutritional security via adequate quality food at affordable prices; covers up to **75% rural** population + up to **50% urban** population(TPDS-subsidised foodgrains) = ~2/3rd of total population; special focus on nutrition for women&children.

5 Green Revolution & Agricultural Revolutions

- White Revolution = speedy ↑ in milk production; Operation Flood started to accelerate it, Father of Operation Flood=**Dr.Verghese Kurien**; largest integrated dairy-dev programme in the world, started by NDDB in **1970**(January 1970 per PYQ) ★
- Yellow Revolution = ↑ oil-seeds production | Blue Revolution = progress in fish production
- 11th Plan(2007-12): agri plan-outlay declined to only 18.5%; agri&allied avg annual GDP growth=3.7% (short of 4% target, but better than 10th Plan's 2.4%)
- Tenth Plan = first plan which did NOT fix crop-production targets
- Green Revolution did NOT cover barley/ragi/minor-millet; confined to HYV mainly rice/wheat/maize/jowar
- National Agriculture Insurance Scheme implemented Oct 1999

DOUBT Green Revolution credit conflict(internal to Lucent, both preserved): p.257 attributes credit to Agriculture Scientist **Dr.Norman Borlaug**(Dr.M.S.Swaminathan's contribution "not less"; its "termed name" attributed to American scientist "Dr.William Gaud" — likely William Gaud, as printed △) — vs p.258 pink box which instead says **M.S.Swaminathan** is "renowned for his leading role in India's Green Revolution" while **Norman E.Borlaug** = "The Father of Green Revolution" in the WORLD. Also: Green Revolution "started in Third Five Year Plan"(Lucent, most positive effect on wheat, 500% crop-production increase as printed) — conflicts w/ Parmar Ch.10's placement in the Plan Holiday(1966-69); see Batch 7.

6 Land Reforms & Land Holdings

- Land reorganisation measures: (1) **Land ceiling**(max land a farmer can hold, excess=illegal), (2) **Chakbandi**(aggregating divided/broken land parcels)
- Land-holding classification: Marginal(<1 hectare), Small(1-4 hectares), Large(>4 hectares)
- Chakbandi first implemented in India in **1920, in Baroda**

7 Agriculture Finance & NABARD

- Unorganised finance sources: money-lenders, money-dealers, relatives, businessmen, landlords, commission agents | Organised sources: Co-operative Committees, Co-operative Banks, Commercial Banks, RRBs, the Government
- Co-operative Credit Organisation started first in **1904**; Primary Co-operative Committees=short-period credit; State Co-operative Agriculture&Rural Dev Banks=long-period credit
- Land Development Bank established **1919**(as Land Mortgage Bank); provides long-term loans

HOOK NABARD → apex institution of Rural Credit; established **12 July 1982** by merging RBI's Agriculture Credit Dept + Agricultural Refinance&Development Corporation(ARDC); based on **Shivraman/Sivaraman Committee** recommendations △. Authorised share capital: ₹500cr(original) → ₹5,000cr(w.e.f.1Feb2001) → ₹30,000cr(NABARD Amendment Bill 2017, passed January 2018; also enabled RBI's exit from NABARD shareholding).

8 Green Revolution — Regional Disparity & PYQ Deep-Dive

- Green Revolution regional disparity ★: HYV/fertilizer/irrigation access initially limited to well-irrigated regions(Punjab/Haryana/Western UP) → rapid growth there, low benefit in rain-fed/less-irrigated areas → regional imbalance

- A-R ★: Green Revolution led to regional imbalances(TRUE) — because HYV tech+irrigation initially available only in select states(Reason TRUE, correct explanation)
- 1st phase(mid1960s-mid1970s) effectiveness ★: highly effective in **Punjab/Haryana**(canal/tube-well irrigation, wheat-centric, HYV seeds); largely **INEFFECTIVE in Bihar**(flood-prone, limited irrigation, weak infra, small landholdings); Tamil Nadu's impact increased only in later phases(not entirely ineffective)
- Punjab = first & largest beneficiary state of the Green Revolution ★; Punjab&Haryana → >85% of net sown area under irrigation(Bhakra Nangal canal system+tubewells) ★

TRAP Structural-transformation failure(1950-1990) ★: despite agri's declining GDP share, ~65% of population remained agriculture-dependent by 1990(workforce didn't shift to industry/services as theory predicts) — from ~75%(1950, ~70-75% per another PYQ variant on Independence) to ~65% (1990) after 4 decades of planning.

Verghese Kurien ★: Father of White Revolution in India; led Operation Flood(dairy modernisation, ↑ farmer income, milk self-sufficiency); chief architect of Amul model+Anand pattern — NOT the inventor of milk pasteurisation(that was **Louis Pasteur**, 19th century — tested distractor). Land reforms(post-independence) key equity goal ★: abolition of intermediaries(Zamindari system)+land-ceiling laws → ownership rights to actual cultivators; central principle="**land to the tiller**"(↑ productivity+social justice).

9 Agri Schemes, Missions & Institutions — PYQ Deep-Dive

- Agriculture Infrastructure Fund(AIF) ★ → supports agri-logistics/cold chains/warehouses/primary-processing units; low-interest loans to farmers/FPOs/cooperative societies/agri-entrepreneurs
- AGMARK ★ → certification mark for **agriculture** products(quality/purity); issued under Agricultural Produce(Grading&Marking) Act, 1937
- National Mission on Natural Farming ★ → launched **November2024**; chemical-free/natural farming for **1 crore** farmers; focus=soil health/↓ input costs/↑ farmer income/sustainable agri
- Technology Mission on Oilseeds(TMO) ★ → launched **1986** by Ministry of Agriculture; NDDB given oilseed procurement/processing responsibility(cooperative model on 'Operation Flood' lines) → launched 'Operation Golden Flow'(edible-oil self-reliance, remunerative farmer prices)
- ULPIN(Unique Land Parcel Identification Number) ★ → unique ID for every land parcel in India(TRUE); 14-digit alphanumeric ID(TRUE); tracks ownership/resolves disputes(TRUE); used in BOTH rural AND urban areas(NOT urban-only — a tested FALSE statement)
- Uttar Pradesh ★ → highest proportion of workforce engaged in agriculture(most populous state, fertile Ganga-Yamuna plain, adequate water, majority-rural population)
- GMOs in agriculture ★ → main purpose=improve crop yield+pest resistance+nutritional quality(via DNA changes) — eg Bt Cotton(pest resistance), Golden Rice(Vitamin-A enriched) — NOT eliminating fertilizers/pesticides entirely or lab-only cultivation(distractors)
- Occupational/workforce shift agriculture→industry&services ★ = "occupational diversification"(↓ rural-economy risk, new income sources, structural economic change)
- Fertilizer subsidies(1960-1990) ★ = "double-edged" policy — boosted Green Revolution productivity BUT benefits mainly reaped by richer/large farmers(small farmers under-benefited) → ↑ rural economic inequality
- SMAM(Sub-Mission on Agricultural Mechanization) ★ → FPOs get **75%** cost grant for Kisan Drone demonstrations(accessible tech for small farmers, precision agriculture)

Post-independence(~1947) main occupation ★: Agriculture(~70-75% of population dependent), highest GDP contributor(manufacturing/services still nascent). AMUL milk cooperative ★: established **14 December 1946**, Anand(Gujarat), under guidance of Sardar Vallabhbhai Patel; originally "Kaira District Co-operative Milk Producers' Union Limited"; laid foundation of White Revolution by ending middlemen exploitation.